AI Agent Assignment Submission – AI Learning Path Planner.

💾 SECTION 1: BASIC DETAILS

Name: Pranishraj Nigade
AI Agent Title / Use Case: AI Agent to help learners generate weekly learning paths based on topic, timeline, and preferences.

🧐 SECTION 2: PROBLEM FRAMING

1.1 What problem does your AI Agent solve?
Learners often struggle to break large goals into achievable weekly plans. They feel overwhelmed without structure or milestones.

1.2 Why is this agent useful?
It converts vague learning intentions into a detailed, week-by-week roadmap tailored to individual goals and preferences.

1.3 Who is the target user?
Students, self-learners, or working professionals trying to upskill in a topic like AI, design, or writing.

1.4 What not to include?
This agent does not recommend specific courses or platforms (e.g., no Coursera/YouTube links), nor does it provide quizzes or grade progress.

🛠️ SECTION 3: 4-LAYER PROMPT DESIGN

🔷 *3.1 INPUT UNDERSTANDING*

Prompt:
"You are a learning coach. Ask the user for the topic they want to learn, how many weeks they have, and any learning preferences."

What is this prompt responsible for?
To understand user intent and capture structured context: topic, duration, style.

Example Input + Output:
Input: "I want to learn machine learning in 4 weeks, prefer

hands-on tasks and visuals."
Output: {topic: "machine learning", duration: 4, preferences: "hands-on, visuals"}

🔷 *3.2 STATE TRACKER*

Prompt:
"Remember the topic, number of weeks, and preferences. Reuse this context in every response."

How does this help the agent "remember"?
Maintains continuity across interactions and enables consistent personalization.

Memory technique:
Simulated via variable holding inside Streamlit and passed as prompt context.

🔷 *3.3 TASK PLANNER*

Prompt:
"Break the topic into {duration} weekly milestones. Each week should include:

Core Concepts
Practice
Resources
Reflection. Ensure logical progression and increasing difficulty."

Agent Thinking:
The agent decomposes abstract goals into weekly structured units. Used prompt chaining internally (learning objective → milestones → weekly plans).

🔷 *3.4 OUTPUT GENERATOR*

Prompt:
"Format the learning path week-by-week using markdown:

Week 1

Core Concepts:
Practice:
Resources:
Reflection: Add motivational tone and make output concise but rich."

Special Behavior:
Used markdown formatting for clarity. Added friendly tone ("You got this!").

🔍 SECTION 4: CHATGPT EXPLORATION LOG

Attempt	Prompt Area	What Happened	What You Changed	Why You Changed It
1	Task Planner	Output lacked progression	Added "increasing difficulty" logic	To simulate structured learning
2	Output Generator	Too verbose	Trimmed text and used markdown	Improved readability
3	Input Understanding	Failed to parse vague input	Added examples and guided questions	Better user input extraction

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🧪 SECTION 5: OUTPUT TESTS

Test 1: Normal Input
Input: "Learn Python in 3 weeks. Prefer real-world tasks."
Output: A 3-week plan with structured tasks and reflection.

Test 2: Vague Input
Input: "I want to learn stuff about AI."
Output: Clarified user needs with a follow-up and generated a starter plan.

Test 3: Empty Input
Input: ""
Output:"Could you tell me the topic, duration, and style you prefer for learning?"

🔄 SECTION 6: REFLECTION

6.1 Hardest Part:
Simulating memory across user input and managing prompt context inside Gemini without system message chaining.

6.2 Most Enjoyed:
Seeing how Gemini could convert basic input into structured weekly learning blueprints.

6.3 Improvements with Time:
Add quiz recommendations, export to calendar integration, or create personalized reminders.

6.4 Learning About Prompt Design:
Prompt structure massively affects model coherence. "Roles" and markdown formatting helped guide the LLM output clearly.

6.5 When Stuck:
Asked ChatGPT for prompt restructuring tips and compared Gemini vs ChatGPT behavior on same prompt.

🧠 SECTION 7: HACK VALUE

Integrated Google Gemini using google-generativeai
Deployed the agent via Streamlit with secure API key handling
Markdown-formatted learning plans with clear structure and motivational tone
Explored edge cases and used fallback prompts
